

Analysing control structures

Jarmo Hurri

Introduction

- you are given a program
- what does it do?
- what should it do?
- does it do what it should do?

Example 1

Consider the algorithm below represented in pseudocode (simplified programming language). The input consists of distinct (not equal) integers A, B and C.

1. Implement the pseudocode in Python.
2. What problem does the algorithm seem to solve?
3. Show, for one possible relationship between the values of A and B and C, that the algorithm solves the problem.

$AB = A - B$

$AC = A - C$

$BC = B - C$

if $AB * BC > 0$ then

$RESULT = B$

else if $AB * AC < 0$ then

$RESULT = A$

else

$RESULT = C$

end if

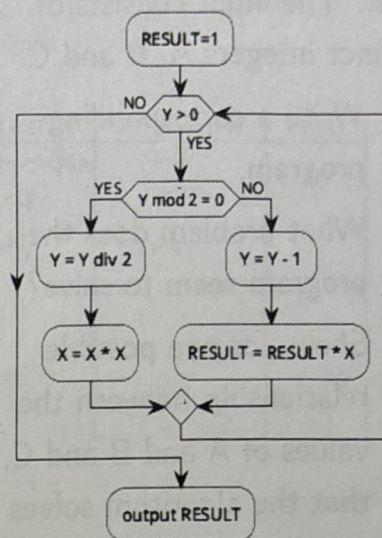
output $RESULT$

Example continues

Example 2

Consider the flowchart to the right (slightly different from IBO notation). The inputs to this program are integers X and $Y \geq 0$.

- Trace the result of this algorithm when $X=2$ and $Y=5$ using the table on next page.
- Write a corresponding program.
- What is the purpose of this program?



Example continues

RESULT	X	Y	Y > 0	Y mod 2 = 0	output
1x2	2	5	YES	NO	
2	2	4	YES	YES	
2	4	2	YES	YES	
2x16=32	16	1	YES	NO	
	16	0	NO		32

Homework 1

Consider the algorithm on the right. The input consists of distinct integers A, B and C.

1. Write a corresponding program.
2. What problem does the program seem to solve?
3. Show, for one possible relationship between the values of A and B and C, that the algorithm solves the problem.

```
RESULT = A
```

```
if B > RESULT then
```

```
    RESULT = B
```

```
end if
```

```
if C > RESULT then
```

```
    RESULT = C
```

```
end if
```

```
output RESULT
```

Homework 2

Consider the function `fun()` below. $\leftrightarrow$

1. Trace the call `fun(k)` when `k=[7, 3, -1, -5, -8, -12]` (see next page).
2. Confirm your trace by adding calls to `print()` inside the function.
3. Call the function with different parameter values. What is the purpose of this function?

```
def fun(x):  
    flag = True  
    i = 2  
    while flag and i < len(x):  
        if x[i] - x[i-1] != x[i-1] - x[i-2]:  
            flag = False  
        else:  
            i += 1  
  
    return flag
```

Homework continues

$$x = [7, 3, -1, -5, -8, -12]$$

2 3 4

flag	i	len(x)	flag and i < len(x)	x[i] - x[i-1]	x[i-1] - x[i-2]	fun(k)
T	2	6	T T - T	-4	-4	
T	3	6	T T - T	-4	-4	
F	4	6	T T - T	-3	-4	

Homework 3 [IB 2016, adapted]

Trace the program, for $N=139$ and $L=3$, in the trace table given on the next page.

```

d = N // L
z = 1
b = False
while z < L:
    d = d // L
    z = z + 1
    b = not b

```

$$139 \div 3 = \approx 46 \quad d = 46$$

$$z = 1$$

```

if d != 0 and b:
    print(d, b)
else:
    print(z, not b)

```

Homework continues

$N=15$ $L=3$

d	z	b	$z < L$	output
46	1	F	T	
46%15	2	T	T	
15%3	3	F	F	3, T.

Consider the function `fun()` below. →

1. Trace the call `fun(k)` when $k = [7, 6, -1, -5, -8, -12]$

(see next page)

2. Confirm your trace by running

adding calls to `fun()` in a script

`print()` inside the function

3. Call the function with

different parameter

values. What is the

purpose of this

function?